PFC AI Design Agent

Supplier Data Integration & Control Theory Foundation

How supplier data is sourced, validated, and documented · How the control theory foundation is structured for all audiences

Status: Discussion — No source code changes

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This document addresses two areas identified as needing explicit treatment: (1) how supplier-provided data — material curves, IC parameters, application note constants — is integrated into the report with full traceability; (2) how the control theory foundation is structured so that every reader, from a first-year student to a control theory professor, follows the entire design reasoning from first principles to final component values.

Part 1 — Supplier Data: Integration into the Report

Every number in the report that comes from a supplier — whether a datasheet, an application note, or a measured curve — must be formally sourced and validated. This is not optional. A report that says "a = 86.7" without showing where 86.7 comes from cannot be reviewed, audited, or reproduced.

✓ AGREED Supplier Data is a First-Class Section in Every Chapter That Uses It

Chapters 3, 4, and 6 each have a dedicated "Data Sources and Model Validation" sub-section that appears before any calculation that uses supplier data. Every parameter is sourced, its physical meaning is explained, and it is validated against at least one known reference point from the datasheet.

1.1 The Four Categories of Supplier Data

Category	Examples	Where Used	Validation Method
Magnetic material curves	B-H curve, permeability vs H, core loss density vs Bac and f	Chapters 3, 4	Fit equation evaluated at two or more datasheet reference points — ✓ check
Magnetic material constants	Steinmetz a, b, c; AL nominal and tolerance; Ae, Le, Ve, Wa	Chapters 3, 4	Model output vs. datasheet loss density table at specific B and f
Thermal data	Core temperature rating, surface emissivity, thermal resistance Rth	Chapter 4	ΔT model cross-checked against manufacturer thermal rating at full rated loss
Control IC parameters	GMI, GMV, VRAMP, VREF; R_M, K_RLPK, K_RM; limits and operating windows	Chapter 6	Three-path G_MOD verification (A/B/C ratio = 1.000 proves internal consistency)
App note equations	AN4165-D Eq. 39 (R_LS), Eq. 40 (R_GC), Eq. 64 (C_SS); SLVA662 Method B	Chapter 6	Result within stated valid range; compared against alternative sizing method

1.2 The DATA SOURCE Box

Every supplier parameter that appears in a calculation is preceded by a "DATA SOURCE" box — visually distinct with a gold left border. It states: the parameter name, the value, the source document and section, the physical meaning, and the validation check.

DATA SOURCE EDGE 60µ Steinmetz Core-Loss Constants

a = 86.7, b = 2.106, c = 1.444 Source: MAGNETICS EDGE material datasheet, core-loss density table. Physical meaning: P_core [mW/cm^3] = a · Bac,pk[T]^b · f[kHz]^c. The exponent b governs how sensitively loss responds to flux swing; c governs frequency sensitivity. For EDGE material, loss grows faster with frequency (c = 1.444) than with flux (b = 2.106) — this favours lower switching frequencies when core loss dominates. Validation: a · 0.1^2.106 · 100^1.444 = 86.7 × 0.00617 × 983 = 525.1 mW/cm^3 ≈ 525 mW/cm^3 (datasheet max @ 0.1 T, 100 kHz) ✓

DATA SOURCE EDGE 60μ Permeability vs DC Bias Model

μ_rel(H) = 1 / (0.01 + 9.202×$10^{■1■}$ · H[Oe]^3.044) Source: MAGNETICS EDGE 60μ material curve, catalog, fitted to two anchor points. Physical meaning: μ_rel is the fraction of initial permeability remaining at DC bias H. As H increases (more DC current through N turns), permeability drops — this is the soft-saturation characteristic of powder cores. Unlike ferrite, the drop is gradual: at 130 Oe μ_rel = 80%, at 205 Oe μ_rel = 50%. Validation: H = 130 Oe → μ_rel = 1/(0.01+9.202e-10×130^3.044) = 80.0% ✓ H = 205 Oe → μ_rel = 1/(0.01+9.202e-10×205^3.044) = 50.0% ✓

DATA SOURCE FAN9672 Internal Control Parameters

G_MI = 88 μS (IEA OTA transconductance) G_MV = 100 μS (VEA OTA transconductance) V_RAMP = 5.0 V (PWM ramp amplitude) V_REF = 2.5 V (internal voltage reference) K_RM = 6000 (gain-modulator bias constant) K_RLPK = 2.465 (ripple-peak detector scale factor) Source: FAN9672-D datasheet Table 1; AND9925-D Table 1. Physical meaning: G_MI converts the error-voltage on the IEA pin to an output current that flows through the IEA compensation network (R_IC, C1, C2). The ratio G_MI/V_RAMP = 88μS/5V = 17.6 μA/V is the net current-loop forward gain from IEA pin voltage to duty-cycle perturbation. Validation: G_MOD three-path check — paths A, B, C agree at 1.0000 at all operating points — proves G_MI, K_RM, and K_RLPK are mutually consistent ✓

1.3 Data Source Section Structure in Each Chapter

Every chapter that uses supplier data has a dedicated sub-section at its start, before any calculation. The sub-section title and location are fixed:

Chapter	Sub-section	Contents
Chapter 3 PFC Inductor Sizing	3.3 Core Material Selection 3.3.3 Material model — constants and validation 3.3.4 Permeability vs bias — model and validation 3.3.5 Core loss density curves (50/70/100 kHz)	Steinmetz constants, fitted bias model, loss density curves from datasheet — all validated before use.
Chapter 4 PFC Inductor Performance Analysis	4.0 Core Thermal Data 4.0.1 Temperature ratings (core, winding, coating) 4.0.2 Surface-area thermal model — source and limits	Maximum temperature ratings from core and wire datasheets. SA model calibrated against published thermal measurements.
Chapter 6 Control Scheme	6.0 Controller IC Model 6.0.1 Internal parameters table 6.0.2 Physical meaning of each parameter 6.0.3 Application note equations used 6.0.4 G_MOD self-audit	Every FAN9672 internal constant sourced to datasheet section. App note equations cited by document number and equation number.

1.4 Curve Data — How Graphs from Datasheets Become Report Figures

When a supplier provides a curve (BH curve, core loss family, impedance vs frequency), the report does not simply reproduce the datasheet image. Instead it shows:

- **The fitted analytical model** — the equation we use in calculations
- **Two validation points** — the model evaluated at two datasheet reference points with ✓
- **The operating point marked** — where this design sits on the curve
- **The safety margin shown** — how far the operating point is from the limit (e.g. permeability knee)

INSIGHT Why We Never Use Curves Directly in Calculations

A curve image cannot be computed with. The design tool needs an equation. The equation must be validated against the curve before use. The validation gives the report reviewer confidence that the equation accurately represents the material at this operating point. Without this step, the design could be based on a misread curve or a model that diverges from the datasheet data at the design operating point.

Part 2 — Control Theory Foundation (Section 6.0)

Before any compensator component value appears, Section 6.0 builds the complete conceptual and mathematical foundation. A student reading this section for the first time will understand why the compensator is needed and what each element does. A control theory professor reading it will find every transfer function derived from circuit equations with no steps skipped.

✓ AGREED Section 6.0 — Control Theory Foundation Precedes All Calculations

A dedicated Section 6.0 is added to Chapter 6 before any component sizing. It covers: the physical block diagram, each block's Laplace-domain transfer function derived from circuit equations, the open-loop Bode plot without compensation (showing instability), what it means to close each loop, pole-zero placement theory, and the step-by-step loop closure sequence.

2.1 Section 6.0 — Complete Outline

6.0.1 The Physical Circuit and What the Controller Must Do

Block diagram of the full PFC power stage. What the controller observes (V_bus, I_inductor) and what it controls (duty cycle D). Why sinusoidal input current requires the duty cycle to vary at line frequency.

6.0.2 Small-Signal Linearisation — Why We Use Laplace

What "small-signal" means: we perturb the operating point by a small amount and observe the response. Why non-linear switching converters can be treated as linear systems around an operating point. Introduction to $s = j\omega$ notation for AC analysis.

6.0.3 Block 1 — The Duty-to-Current Plant G_id(s)

Physical derivation from volt-second balance on the inductor. Transfer function in standard form: one pole, one RHP zero. Bode plot shape: -20 dB/dec slope, phase starting at -90°, falling to -270° above the RHP zero. Why the RHP zero is dangerous: it adds phase lag like a pole but gain like a zero.

6.0.4 Block 2 — The Current-to-Voltage Plant G_vp(s)

Physical derivation: the controlled inductor current charges C_O through R_L. Transfer function: one pole from C_O×R_L, one zero from C_O×ESR. The voltage-path RHP zero: uses L_eq = L_ϕ/2, not L_ϕ. Bode plot shape: flat to f_0, -20 dB/dec above, +20 dB/dec at ESR zero.

6.0.5 Block 3 — Current Sense Filter H_cs(s)

Simple RC low-pass: R_F = 2 kΩ, C_F = 470 pF. Transfer function: single pole at $f_{RC} = 1/(2\pi \times R_F \times C_F) \approx 169$ kHz. Effect on loop: negligible below 50 kHz, rolls off gain near switching frequency. Physical purpose: reject switching-frequency noise from current sense.

6.0.6 Block 4 — PWM Ramp Normalisation (1/V_RAMP)

The comparator converts V_IEA to duty cycle. Gain = 1/V_RAMP = 1/5 = 0.2 V^{-1}. This is a pure scalar — no frequency dependence. Physical meaning: 1 V change in V_IEA moves duty cycle by 20%.

6.0.7 Block 5 — Gain Modulator G_MOD

GMOD scales the current command with V_LPK (line peak voltage). Purpose: without GMOD, loop gain would vary 9:1 over the line range ($90^2 : 264^2$). GMOD makes the crossover frequency nearly independent of Vin. Transfer function: frequency-independent scalar at the IEA sizing corner.

6.0.8 The Uncompensated Loop — Open-Loop Bode Plot

Cascade all five blocks: T_unc(s) = G_id × H_cs × (1/V_RAMP) × G_MOD. Plot the Bode diagram. Key observation: at ANY crossover above 100 Hz, the phase is already below -180°. The loop is unstable without compensation. This motivates the Type II compensator: its zero must push phase above -135° (45° PM target) at the chosen crossover frequency.

6.0.9 What Loop Gain = 1 (0 dB) Means

At the crossover frequency ω_ci, the loop must have unity gain: |T_unc(jω_ci)| × |G_comp(jω_ci)| = 1. This is the fundamental sizing equation for R_IC. Physical interpretation: at ω_ci the signal goes around the loop and returns with the same amplitude — neither growing nor decaying. Phase margin determines how quickly it decays just above this frequency.

6.0.10 Phase Margin — Physical Meaning

PM is the angle between the loop phase at crossover and -180°. PM = 45° → damping ratio ζ = 0.38 → step response overshoot ≈ 26%. PM = 60° → ζ = 0.57 → overshoot ≈ 9%. PM = 0° → ζ = 0 → sustained oscillation. PM < 0° → the system is unstable — perturbations grow. The current loop targets PM ≥ 45°; the voltage loop targets PM ≥ 60° because it must also reject 120 Hz ripple without distorting input current.

6.0.11 Type II Compensator — Pole-Zero Placement

The Type II OTA has: one integrator (→∞ gain at DC → zero steady-state error), one zero at f_z (phase boost near crossover), one pole at f_p (attenuates gain above crossover — rejects switching noise). Placement rules: f_z = f_ci / 5 (zero well below crossover for maximum phase boost), f_p = f_sw / 4 (pole at Nyquist quarter — rejects noise without reducing PM). Result: +70° to +80° of phase boost at f_ci. Transfer function H_comp(s) = G_MI × R_IC × (1 + s/ω_z) / (1 + s/ω_p).

6.0.12 Type III Compensator — When and Why

Type III adds a second zero-pole pair. It is needed when the uncompensated phase at f_cv is below -90° — meaning the single zero of Type II cannot provide enough phase boost for PM ≥ 60°. This occurs when f_cv approaches the LC resonance (both poles compound). The decision criterion: compute ∠T_v,base(jω_cv). If phase > -90°: Type II sufficient. If phase < -90°: Type III required. The additional zero-pole pair (z2, p2) provides +45° to +90° extra boost following SLVA662 Method B placement rules.

6.0.13 Closing the Inner Current Loop First

The inner loop is closed before the outer loop is designed. Physical reason: a closed inner loop converts the inductor from a passive component into a controlled current source. The voltage plant $G_{vp}(s)$ then sees a current input (not a voltage input) — its transfer function simplifies and the inner-path RHP zero is no longer in the voltage loop. Closed-loop inner response: $T_{i\_cl} = T_i / (1 + T_i) \approx 1$ at low frequency. This approximation holds for all frequencies below the inner crossover f_{ci}.

6.0.14 The Outer Voltage Loop — What It Sees After Inner Loop Closure

With the inner loop closed, the voltage plant input is the current command I_{ref}. The outer loop drives a current source feeding C_O: dominant pole at f_{cv}. The voltage-path RHP zero (using $L_{eq} = L_\phi/2$) sets the hard bandwidth ceiling: $f_{cv,max} = f_{RHP\_v} / 5$. The 120 Hz rejection requirement adds a second constraint: the loop gain at 120 Hz must be ≥ 20 dB. This prevents the 120 Hz voltage ripple from modulating the current reference and distorting the input current waveform.

6.0.15 Step-by-Step Loop Closure Sequence

Step 1: Size R_{CS} — set the current sense scale factor. Step 2: Verify G_{MOD} — three-path self-audit. Step 3: Design current loop — choose f_{ci}, size R_{IC}/C1/C2, verify PM at all corners. Step 4: Close the inner loop analytically. Step 5: Compute voltage plant with inner loop closed. Step 6: Choose f_{cv} subject to $f_{RHP}/5$ ceiling and 120 Hz rejection floor. Step 7: Decide Type II or Type III from uncompensated voltage loop phase. Step 8: Size voltage compensator — R2, C1, C3 (or R2, R3, C1, C2, C3). Step 9: Verify both loops at all 9 operating points — stability scorecard.

Part 3 — Transfer Function Format: Derivation Examples

Every transfer function in Section 6.0 is derived from circuit equations, not just stated. The derivation uses the same equation format agreed in the Documentation Standards. The following examples show exactly how they appear.

3.1 Example — G_id(s): Duty-to-Inductor-Current Transfer Function

CONTROL THEORY What G_id(s) Represents Physically

G_id(s) answers the question: "If I increase the duty cycle by a small amount δD, how much does the average inductor current change?" This is the most critical transfer function because it is the plant that the inner current loop must control. Its RHP zero means increasing D first reduces energy delivered to the output before the inductor current builds up — the hallmark of the boost converter's non-minimum-phase behaviour.

CONCEPT Deriving G_id from Volt-Second Balance

We apply a small perturbation δD to the steady-state duty cycle D■. The inductor volt-second balance gives us the inductor current response. We Laplace-transform the linearised equations to find the transfer function G_id(s) = δI_L(s) / δD(s). The result has the standard boost converter form: one pole from the LC resonance and one right-half-plane zero from the boost topology.

Derivation steps:

Step 1 Volt-second balance on the inductor (per phase, CCM): *Eq. 6.1*

$$V_{in} \cdot D + (V_{in} - V_{out}) \cdot (1 - D) = 0 \text{ (steady state)}$$

$$\rightarrow V_{in} = V_{out} \cdot (1 - D) = V_{out} \cdot D' \text{ [where } D' = 1 - D\text{]}$$

$$\mathbf{V_{out} = V_{in} / D' \rightarrow \text{boost relationship confirmed}}$$

Step 2 Linearise around operating point (D■, I_L0, V_out0): *Eq. 6.2*

$$L \cdot d\delta I_L/dt = \delta V_{in} \cdot D\blacksquare + V_{in0} \cdot \delta D - \delta V_{out} \cdot D'\blacksquare + V_{out0} \cdot \delta D$$

$$\text{Taking Laplace } (\delta V_{in} = 0, \text{ output voltage held by } C\_O):$$

$$\mathbf{sL \cdot \delta I_L(s) = (V_{in0} + V_{out0}) \cdot \delta D(s) - D'\blacksquare \cdot \delta V_{out}(s)}$$

Step 3 Combine with output capacitor equation and load: *Eq. 6.3*

$$G_{id}(s) = \delta I_L(s)/\delta D(s) = (V_{out}/L) \cdot (1 - s \cdot L \cdot D'^2/R_L) / (s^2 LC + s \cdot L/R_L + D'^2)$$

$$\equiv K_f \cdot (1 + s/\omega_z) / (s^2 + s \cdot \omega_n/Q + \omega_n^2)$$

$$\mathbf{\text{RHP zero: } \omega_z = R_L \cdot D'^2 / L \text{ (positive zero, negative phase)}}$$

INSIGHT The RHP Zero Frequency Determines Everything

$f\_RHPz = R\_L \cdot D'^2 / (2\pi \cdot L)$. At 90 Vac full load: D' = Vin,pk/Vout = 0.324, $R\_L = 394^2/1700 = 91.3\ \Omega$, L = 235 µH → f_RHPz = 6.45 kHz. The inner current loop crossover f_ci = 8.0 kHz is ABOVE this RHP zero. This appears dangerous — but it is acceptable because the RHP zero of G_id affects the current loop, and the current loop needs less phase margin (45° target) than the voltage loop. The voltage-loop RHP zero uses L_eq = L/2 → f_RHPz_v = 12.9 kHz, and the voltage loop crossover f_cv = 17 Hz is safely below f_RHPz_v/5 = 2.58 kHz by more than 100:1.

3.2 Example — Open-Loop Bode Plot Description (Without Compensator)

CONTROL THEORY What Happens Without a Compensator

If you close the current loop with G_id, H_cs, and the PWM ramp but no compensator — just a resistor R_IC with no capacitors — the uncompensated loop gain at 8 kHz is $|T\_unc(j\cdot 2\pi\cdot 8000)| \approx 0.12$ (-18 dB) and the phase is -147°. Phase margin = 180° - 147° = 33°. Marginally stable, oscillatory response. The compensator's zero at 1.75 kHz boosts phase to +68° at 8 kHz, giving PM = 63°. Without it, the design would ring on any load step.

The uncompensated loop Bode plot shows three distinct regions:

- **Below f_0 (LC resonance ≈ 72 Hz):** Plant rolls at -20 dB/dec (inductor pole dominant). Phase is -90°. This is the ideal operating region for the voltage loop.
- **Near f_0:** The LC double-pole adds another -40 dB/dec (total -60 dB/dec) and -180° of phase in a narrow band. This is where the voltage loop resonance sits and why voltage-loop bandwidth must stay well below f_0.
- **Above f_0 up to f_ci:** Rolls at -20 dB/dec, phase recovering from the resonance. The current loop crossover (8 kHz) sits in this region — one pole dominant, predictable phase, Phase margin achievable with a single zero.
- **Above f_RHPz (6.45 kHz):** RHP zero adds +20 dB/dec to magnitude (making gain appear higher) but adds -90° of phase (making the system less stable). This is the region that limits current-loop crossover to ≤ f_RHPz × 2 or so.

3.3 The Step-by-Step Loop Closure Sequence in the Report

Section 6.0.15 describes the closure sequence. In the body of Chapter 6, each of the 9 steps in the closure sequence becomes a numbered calculation section (6.3 through 6.8) using the full equation format with CONCEPT, THEORY, PITFALL, DECISION, and Interpretation blocks agreed in the Documentation Standards.

Closure Step	Section in Chapter 6	Key equation(s)	Pass criterion
1 — Size R_CS	6.3	$R\_CS = V\_AC^2 \cdot G\_max \cdot R\_M / (R\_IAC \cdot P\_ch)$ [AN4165-D Eq. 31] R_CS < V_EA_window from AND9925-D Eq. 11	Both methods agree; V_EA implied in 4–5 V window

2 — Verify G_MOD	6.3.3	Path A = $K\_RM \cdot R\_IAC/(8 \cdot K^2\_RLPK \cdot R^2\_RLPK)$ Path B = $R\_CS \cdot P\_ch/V\_EA\_eff$ Path C = $K\_max \cdot I\_OUT/V\_EA\_eff$	A/B = 1.000 at all operating points
3 — Current loop	6.4	f_ci = f_sw/8; f_z = f_ci/5; f_p = f_sw/4 R_IC = $1/(\lvert T\_unc(j\omega\_ci)\rvert \cdot G\_MI \cdot \kappa)$	PM ≥ 45° at all 9 pts
4 — Close inner loop	6.0.13	T_i_cl = T_i/(1+T_i) G_icl ≈ 1 (for f << f_ci)	Confirmed analytically
5 — Voltage loop plant	6.5	T_v,base = G_icl · G_vp · G_MOD f_RHP_v = $R\_L \cdot D'^2/(2\pi \cdot L\_eq)$	f_cv < f_RHP_v / 5
6 — Type II or III?	6.5.1	Compute $\angle T\_v,base(j\omega\_cv)$ If phase > -90°: Type II. If < -90°: Type III	Criteria met for selected type
7 — Size voltage comp.	6.5.3	R2 = G_req / (G_MV · H_div · κ) SLVA662 Method B pole/zero placement	PM ≥ 60°, 120 Hz rej ≥ 20 dB
8 — Stability scorecard	6.6	Both loops swept at all 9 operating points PM_i, PM_v, GM_i, GM_v, 120 Hz rejection	All 9 operating points PASS

Part 4 — How Control Theory Blocks Appear in Chapter 6

The following shows the complete structure of Chapter 6 as updated with the agreed Section 6.0 control theory foundation. The entire chapter is a continuous narrative from first principles to final component values.

Section	Title	Content Summary	New in this plan?
6.0	Control Theory Foundation	15 sub-sections: physical block diagram, all 5 Laplace transfer functions derived, open-loop instability demonstration, PM explanation, pole-zero placement theory, loop closure sequence.	NEW — fully agreed in Part 2
6.1	IC Model and Parameter Sources	DATA SOURCE boxes for all FAN9672 internal constants. Pin-by-pin table. Three-path G_MOD self-audit.	NEW section — agreed in Part 1
6.2	Plant Analysis at All Operating Points	f_0, f_esr, f_RHP (current path), f_RHP (voltage path) at all 9 pts. Bandwidth target derivation and rationale.	Existed, enhanced
6.3	R_CS and G_MOD Verification	Both AN4165-D methods, V_EA window, G_MOD three-path audit. Full CONCEPT/PITFALL/DECISION treatment.	Existed, enhanced
6.4	Current Loop Design	Uncompensated Bode → PM problem shown → Type II sizing → achieved margins → Bode plot.	Existed, enhanced
6.5	Voltage Loop Design	Inner loop closed → voltage plant → Type II/III decision criterion → SLVA662 Method B → achieved margins → 120 Hz rejection → Bode plot.	Existed, enhanced
6.6	Stability Scorecard	All 9 operating points. All four margins. Worst-case row highlighted. Interpretation: what each margin means in practice.	Existed, enhanced
6.7	Soft-Start and Protection	C_SS, ILIMIT, ILIMIT2 sizing from AN4165-D with full derivation.	Existed, enhanced
6.8	Control Network BOM	All compensator, divider, protection, and soft-start components with function annotation.	Existed, enhanced

Document Status: Both topics in this document are finalised. Supplier data integration applies to Chapters 3, 4, and 6. Control theory foundation is Section 6.0 of Chapter 6. No source code changes have been made.

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